

2.2b: -1.0: The variance derivation is incorrect. The correct dependence on batch size is not properly derived.

2.2c: -0.5: The conclusion about doubling batch size is incorrect. Gradient noise should decrease, not stay unchanged.

2.3b: -0.5: The suggested interventions are only partially convincing and not as targeted as expected.

2.4: -0.5: One example is less precise and shifts away from leakage/evaluation issues toward a more general training bug.

2.5a: -0.5: The parameter-count derivation is incomplete, especially regarding bias terms and clear FC formula presentation.

2.6b: -0.5: The explanation of pooling invariance is vague and does not clearly distinguish stability from exact invariance.

2.6c: -0.5: The counterexample showing max-pooling is not exactly translation-equivariant is not clearly presented.

2.7: -0.5: The explanation mixes residual connections with U-Net-style skip connections, which makes the argument less precise.

2.9a: -0.5: The general forward-pass expression for the RNN is not written correctly.

2.9b: -0.5: The BPTT recurrence for $\partial L / \partial h_t$ is incorrect.

2.9c: -0.5: The explanation of vanishing/exploding gradients is mathematically inaccurate in several places.

Score

19.5 / 33 - D+

Feedback Date

Mar 23, 2026 10:47 AM

Assignment

Assignment 2

Submission ID	Submission(s)	Date Submitted ▼
7943235	  ELG5218 - Ass 2 Prob 2.docx (43.82 KB)	Feb 28, 2026 1:57 AM
	  SVI assignment problem 1 report.pdf (2.47 MB)	
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7933948	  ELG5218 - Ass 2 Prob 2.docx (43.62 KB)	Feb 26, 2026 6:49 PM
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7907075	  SVI_assignment_problem_1_report.pdf (2.47 MB)	Feb 22, 2026 6:44 PM
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